UNIT-IV: Routing Algorithms & Internet Working

- **Routing Algorithms:**

The Optimality Principle – Shortest Path – Flooding – Distance Vector – Link State – Hierarchical.

- **Congestion Control Algorithms:**

General Principles of Congestion Control – Congestion Prevention Policies – Approaches to Congestion Control – Traffic-Aware Routing – Admission Control – Traffic Throttling – Load Shedding.

- **Traffic Control Algorithm:**

Leaky Bucket & Token Bucket.

- **Internet Working:**

How Networks Differ – How Networks Can Be Connected – Tunneling – Internetwork Routing – Fragmentation – Network Layer in the Internet – IP Protocols – IP Version 4 Protocol – IPv4 Header Format – IP Addresses – Classful Addressing – CIDR – Subnets – IP Version 6 – The Main IPv6 Header – Transition from IPv4 to IPv6 – Comparison of IPv4 & IPv6.

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4.1 Routing Algorithms:

4.1.1 The Optimality Principle:

Introduction to Routing

What is Routing?

Routing is the process of selecting the best path through a network for transferring data packets from a source node to a destination node.

For example:

Computer A

|

v

Router R1 ---- Router R2

| |

v v

Router R3 ---- Router R4

|

v

Computer B

If A wants to communicate with B, the network must determine:

Which path should the packet take?

This decision is made using a routing algorithm.

Routing vs Forwarding

Routing	Forwarding
Determines the best path	Sends packet to the next hop
Uses routing algorithms	Uses forwarding table
Network-wide decision	Local decision
Example: OSPF, RIP	Router forwards packet through an interface

Simple example

Suppose:

A ---- B ---- D

\ /

\---- C ---/

If the cost of A-B-D = 5 and A-C-D = 8, routing chooses:

A → B → D

because it has lower cost.

- **Optimality Principle**

The optimality principle is a fundamental principle used in routing.

Definition

If router J lies on the optimal path from router I to router K, then the optimal path from J to K must also be part of the optimal path from I to K.

In simple words:

The best route from A to D contains the best route from an intermediate router to D.

Example

Consider:

A ---- B ---- C ---- D

If:

A → B → C → D

is the shortest path from A to D, then:

B → C → D

must be the shortest path from B to D.

Why is it important?

It allows routers to calculate routes efficiently and forms the basis of many shortest-path routing algorithms.

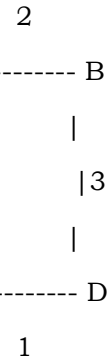
4.1.2: Shortest Path Routing

The objective is to find the **least-cost path** between two nodes.

The "cost" may represent:

- Number of hops
- Delay
- Bandwidth
- Reliability
- Monetary cost
- Congestion
- A combination of these

Example



Possible paths from A to D:

- A → B → D = 2 + 3 = **5**
- A → C → D = 4 + 1 = **5**

Both have equal cost.

Dijkstra's Algorithm

Dijkstra's algorithm is commonly used to find shortest paths.

Dijkstra's algorithm is a famous graph search method used to find the **shortest path** from a single source node to all other nodes in a weighted graph (where the connections between points have numerical values, like distances or travel times). It is the backbone of modern technologies like Google Maps routing.

Here is the step-by-step framework it uses:

1. **Initialization:** Mark your starting node with a distance of **0** and all other nodes with a distance of **infinity** (since you don't know how far they are yet). Mark all nodes as unvisited.
2. **Select the Closest Node:** Look at all unvisited nodes and pick the one with the smallest tentative distance. For the very first step, this will always be your starting node.
3. **Update Neighbours:** For the current node, look at all of its unvisited neighbours. Calculate their distance *through* your current node. If this new distance is shorter than the previously recorded distance, update it.

4. **Mark as Visited:** Once you have checked all neighbours of the current node, mark it as **visited**. A visited node is locked in and its shortest path will not be checked again.
5. **Repeat:** Repeat steps 2 through 4 until every single node in the graph has been visited.

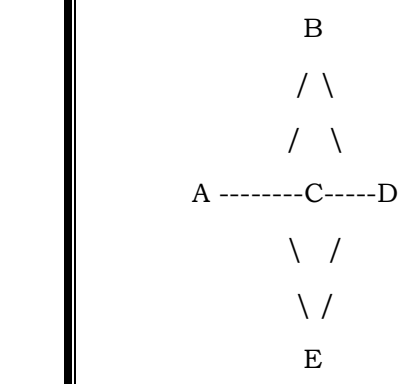
Basic steps

1. Assign distance 0 to source.
2. Assign infinity to all other nodes.
3. Select the unvisited node with the smallest distance.
4. Update distances of its neighbors.
5. Mark the node as visited.
6. Repeat until all required nodes are processed.

4.1.3 Flooding

Definition

Flooding is a routing technique in which every incoming packet is sent through **all outgoing links except the link through which it arrived**.



If A sends a packet to C, C forwards copies to B, D and E.

Advantages

- Very simple
- Highly robust
- Does not require complete routing information
- Useful when reliability is important

Disadvantages

- Generates a large number of duplicate packets
- Wastes bandwidth
- Can cause congestion

Controlling flooding

Techniques include:

- Hop count
- Sequence numbers

- Selective flooding

Simple analogy

Imagine shouting a message in a building and asking **everyone to repeat it to everyone else**.

The message will certainly spread, but there will be enormous duplication.

Distance Vector Routing

Distance Vector routing is a **dynamic routing algorithm**.

Each router maintains a table containing:

Destination	Distance	Next Hop
A	0	—
B	2	B
C	5	B
D	7	C

Each router periodically shares its routing information with its **neighbouring routers**.

Example

A -------- B -------- C

Initially:

- A knows B
- B knows A and C
- C knows B

B tells A that C is 2 hops away.

A can then learn:

A → B → C

4.1.4 Bellman-Ford concept

Distance Vector routing is based on the Bellman-Ford approach.

The basic relationship is:

$$D_x(y) = \min_v \{c(x,v) + D_v(y)\}$$

where:

- $D_x(y)$ = distance from x to y
- $c(x,v)$ = cost from x to neighbor v
- $D_v(y)$ = distance from v to y

Advantages

- Simple

- Easy to implement
- Requires less information

Problems

Count-to-infinity problem is an important problem.

Other issues:

- Slow convergence
- Routing loops
- Incorrect routing information during failures

Example protocol

RIP – Routing Information Protocol uses distance-vector routing.

4.1.5 Link State Routing

In **Link State Routing**, each router builds a complete view of the network topology.

Instead of simply saying:

"My destination is 4 hops away."

the router shares information about its links.

Basic steps

1. Discover neighbors.
2. Measure link cost.
3. Create link-state packet.
4. Flood link-state information.
5. Build network topology.
6. Run shortest-path algorithm.
7. Each router attempts to construct a similar topology map.

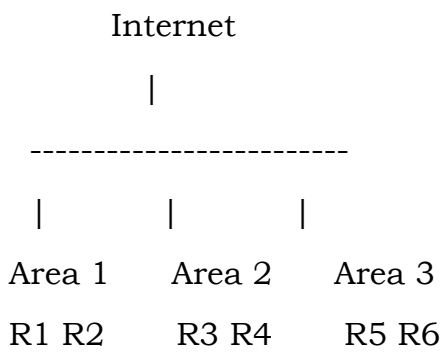
4.1.6 Link State vs Distance Vector

Distance Vector	Link State
Shares routing information with neighbors	Distributes link-state information
Does not maintain complete topology	Maintains topology database
Bellman-Ford	Dijkstra
Slower convergence	Faster convergence
Can suffer from count-to-infinity	Less susceptible
Example: RIP	Example: OSPF

4.1.7 **Hierarchical Routing**

Large networks cannot efficiently maintain detailed routing information about every router.

Therefore, networks are divided into **regions or areas**.



Routers inside an area maintain detailed information about that area.

For other areas, summarized information is used.

Advantages

- Reduces routing table size
- Reduces routing overhead
- Improves scalability
- Faster routing decisions

Real-world example

OSPF supports hierarchical routing using areas.

4.2 Congestion Control Algorithms:

In computer networks, congestion occurs when too much data is sent through the network at the same time, causing delays, packet loss, or even network collapse. This is similar to a traffic jam on a highway - when too many cars try to move at once, everything slows down.

Congestion control is the set of techniques and mechanisms used to prevent, detect, and manage congestion. Its main goal is to maintain smooth data flow, fair bandwidth allocation, and efficient utilization of network resources, ensuring a stable and reliable network performance.

Effects of Congestion Control

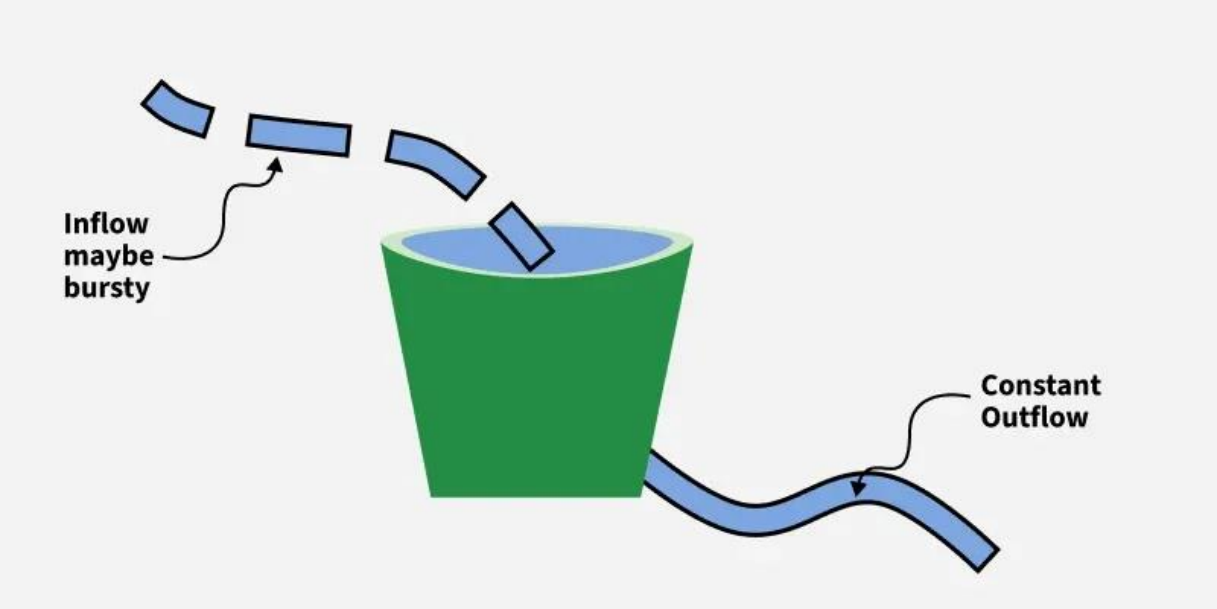
1. **Improved Network Stability:** Prevents overload and keeps the network running smoothly.
2. **Reduced Latency and Packet Loss:** Ensures data is delivered faster with fewer retransmissions.
3. **Enhanced Throughput:** Maximizes the volume of data successfully transferred in a given time.
4. **Fairness in Resource Allocation:** Distributes bandwidth evenly so no user monopolizes resources.
5. **Better User Experience:** Provides faster access to websites, applications, and services.

6. **Prevention of Congestion Collapse:** Avoids severe breakdowns where the network becomes nearly unusable.

Congestion Control Algorithms

1. Leaky Bucket Algorithm

The leaky bucket algorithm controls the flow of packets by sending them out at a constant, fixed rate, regardless of incoming burst traffic. Packets are placed into a bucket (queue), and if the bucket overflows, excess packets are discarded. This ensures smooth traffic but wastes bandwidth during idle times.



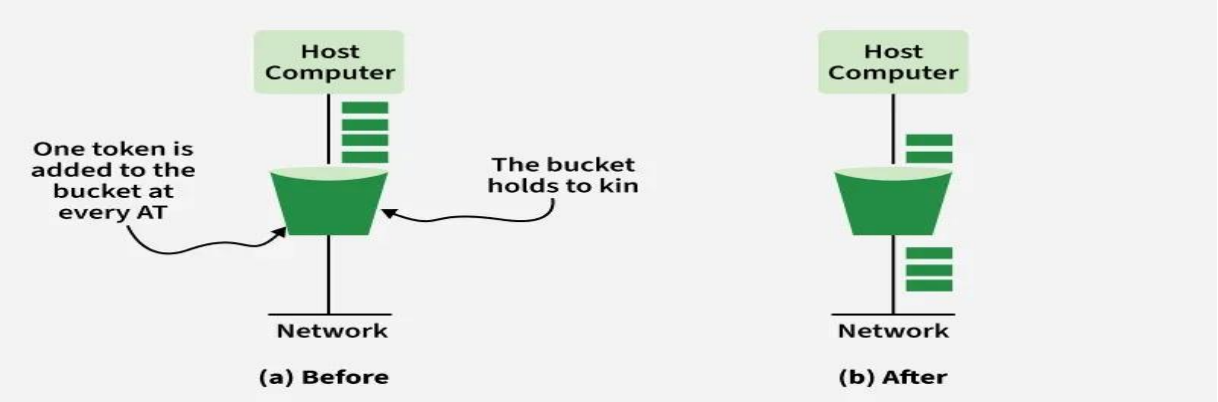
Steps:

- Packets arrive -> placed into the bucket.
- Bucket leaks (transmits) at a constant rate.
- Bursty traffic is smoothed into uniform traffic.

Limitation*: Too rigid, wastes available bandwidth if traffic is bursty.*

2. Token Bucket Algorithm

The token bucket algorithm allows more flexibility by generating tokens at a fixed rate. A packet can only be transmitted if a token is available, and each packet consumes one token. If enough tokens accumulate, burst traffic can be sent quickly, making it better for handling variable traffic patterns without unnecessary drops.



Steps:

- Tokens are added to the bucket at regular intervals.
- Each token permits sending one packet.
- If tokens exist -> packets can be transmitted immediately.
- If no tokens -> packets must wait.

Advantage*: Handles bursty traffic efficiently without unnecessary data loss.*

Leaky Bucket vs Token Bucket

Parameter	Leaky Bucket	Token Bucket
Output	Fixed, constant rate	Variable, allows bursts
Flexibility	Rigid	Flexible
Packet Loss	Possible during bursts	Avoids loss if tokens available

Need of Token Bucket: *The leaky bucket algorithm enforces output pattern at the average rate, no matter how bursty the traffic is. So in order to deal with the bursty traffic we need a flexible algorithm so that the data is not lost.*

Advantages of Congestion Control

- Ensures stable and reliable operation of networks.
- Reduces delays and retransmissions.
- Minimizes data loss.
- Optimizes resource utilization.
- Scales well with growing networks.

- Adapts to changing traffic conditions.

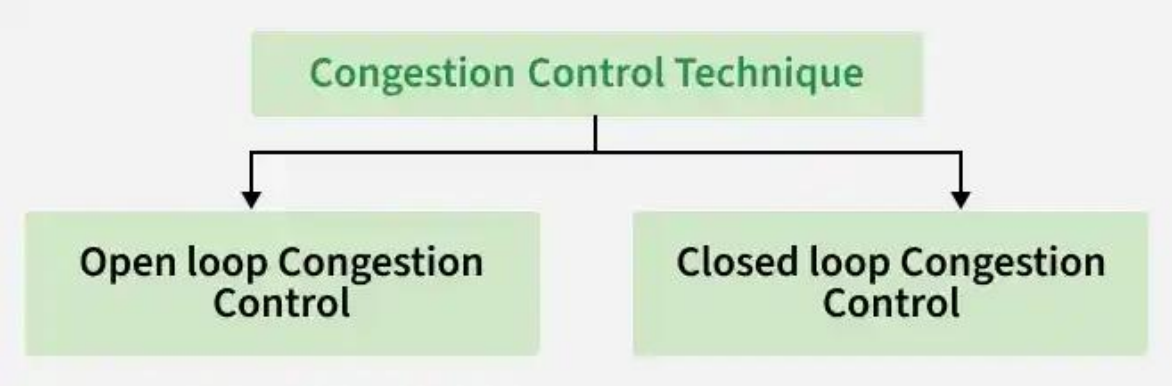
Disadvantages of Congestion Control

- Adds complexity to network design.
- May introduce overhead in processing.
- Can be sensitive to network conditions.
- May struggle with resource prioritization in critical scenarios.
- Effectiveness depends on modern infrastructure.

4.2.1 General Principles of Congestion Control:

Congestion control is a mechanism that controls a sender's data transmission rate so the network remains stable under varying traffic loads.

- It keeps network delay under control by limiting how much data builds up in router queues.
- It reduces jitter by preventing queues from growing and shrinking unpredictably during traffic bursts.
- It lowers packet loss by avoiding buffer overflow in routers and switches.
- It improves fairness by ensuring that multiple connections share available bandwidth more evenly.
- It maintains stable throughput by reducing retransmissions and avoiding repeated overload cycles.



Types of Congestion Control

These techniques are broadly classified into two categories: Open-Loop (preventive) methods and Closed-Loop (reactive) methods.

1. Open-loop congestion control

Open-loop congestion control prevents congestion by using predefined rules and design-time policies, instead of relying on real-time feedback from the network during transmission.

Open Loop Congestion Control Policies

1. Retransmission Policy: Lost or corrupted packets are retransmitted by the sender. To prevent congestion, retransmission timers must be carefully set to avoid excessive retries, which could worsen network load.

2. Window Policy: The type of sliding window protocol affects congestion. In Go-Back-N ARQ, multiple packets may be retransmitted for a single loss, increasing network load. Selective Repeat ARQ is preferred, as it retransmits only the missing packets, reducing unnecessary congestion.

3. Discarding Policy: Routers can selectively discard less important, damaged, or delayed packets to prevent congestion. This is useful in multimedia applications like audio or video streaming, where losing some packets may not noticeably affect quality.

4. Acknowledgment Policy: Acknowledgment traffic adds load to the network. Congestion can be reduced by employing strategies such as sending a single acknowledgement for multiple packets or delaying acknowledgements until a packet is ready to be sent or a timer expires.

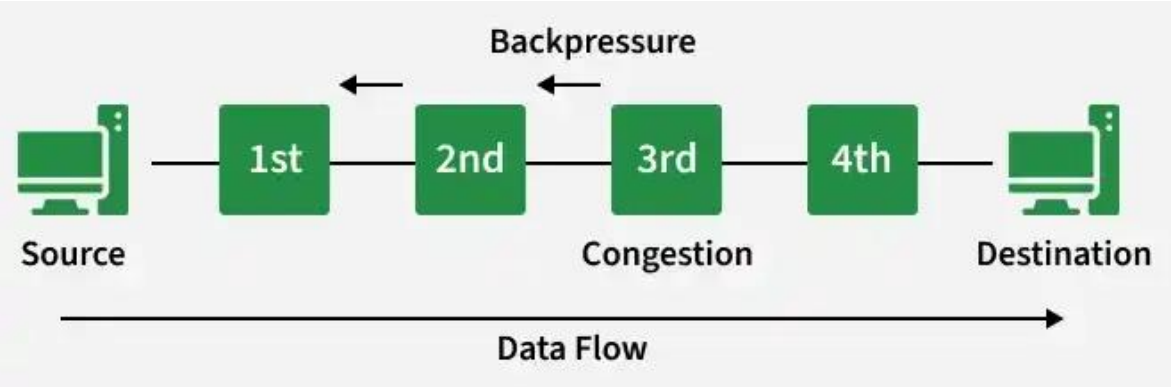
5. Admission Policy: Admission control allows new connections only if sufficient network resources are available. Before establishing a connection, routers or switches check bandwidth and buffer availability. If adding a new flow would risk congestion, the request is denied to protect existing traffic.

6. Traffic Shaping and Rate Control (Additional): Open loop systems may also include traffic shaping, where sources limit the rate of packet transmission, and rate control, ensuring that traffic conforms to agreed limits before entering the network.

2. Closed-loop congestion control

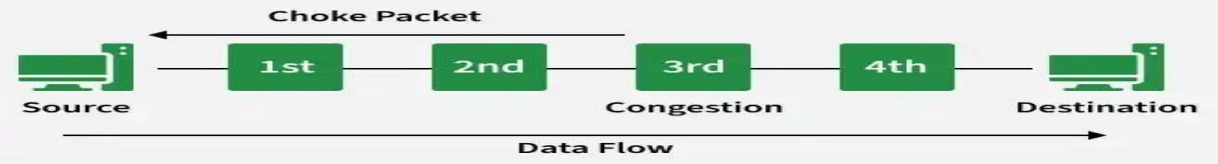
Closed-loop congestion control detects congestion during communication using feedback from the network and then reacts by adjusting the sending rate or network behavior to reduce overload.

1. Backpressure: A congested node temporarily stops accepting packets from its upstream neighbors, slowing traffic back toward the source. This method propagates congestion signals backward through the network. It works primarily in virtual circuit networks, where nodes know their upstream neighbors. For example, if the 3rd node is congested, it stops receiving packets, causing congestion at the 2nd node, which then affects the 1st node and eventually informs the source to slow down.



In above diagram the 3rd node is congested and stops receiving packets as a result 2nd node may be get congested due to slowing down of the output data flow. Similarly 1st node may get congested and inform the source to slow down.

2. Choke Packet Technique: When a router experiences congestion, it sends a special choke packet directly to the source, instructing it to reduce its transmission rate. The router monitors its resource usage and triggers the choke packet when utilization exceeds a threshold. Only the source is notified, while intermediate nodes remain unaware of the congestion.



3. Implicit Signaling: In this method, no explicit congestion message is sent. Instead, the source infers congestion from network behavior, such as delayed or missing acknowledgments, and adjusts its sending rate accordingly.

4. Explicit Signaling: A congested node directly informs the source or destination about congestion by embedding congestion information inside data packets. This differs from choke packets, which are separate control packets.

- **Forward Signaling:** Congestion information is sent to the destination, which then takes measures to handle or avoid congestion.
- **Backward Signaling:** Congestion information is sent to the source, instructing it to slow down its transmission rate.

Network-assisted Congestion Control

Network-assisted congestion control is a method where routers actively support congestion handling by signaling congestion early or controlling queue buildup, instead of leaving everything to the sender's loss-based guessing.

Network-assisted Techniques

1. ECN (Explicit Congestion Notification): ECN allows routers to mark packets instead of dropping them when congestion is starting. The receiver echoes this mark back to the sender, and the sender reduces its sending rate, which helps control congestion without waiting for packet loss to occur.

2. AQM (Active Queue Management): AQM is a queue management approach where routers start dropping or marking packets before the buffer becomes full. This prevents long queues from building up, which reduces queueing delay and makes network performance smoother during peak traffic.

3. CoDel (Controlled Delay): CoDel is an AQM technique that focuses on controlling delay rather than just buffer size. It monitors how long packets stay in the queue, and if the delay remains high for a sustained time, it drops or marks packets to force senders to slow down and bring the queue delay back to a healthy level.

4. Why it is useful: Network-assisted methods reduce the chance of buffer overflow and reduce sudden bursts of packet loss, because congestion is handled earlier through marks or controlled drops.

5. Key benefit for real networks: These methods keep latency-sensitive applications like voice and video more stable, because they prevent queues from becoming excessively long during heavy load.

4.2.2 Congestion prevention policies

(Also known as **open-loop congestion control**) are proactive strategies built into the network design to prevent congestion before it even begins.

Instead of reacting after the network slows down, these policies use strict, pre-defined rules across different layers of the **Computer Network (CN)** architecture to regulate data flow.

Key Congestion Prevention Policies

These policies operate at different stages of data transmission to minimize the risk of a bottleneck:

- **Retransmission Policy**: Dictates how a sender handles timed-out packets. If a sender is too aggressive or uses an inefficient protocol (like Go-Back-N instead of Selective Repeat), it will flood the network with unnecessary duplicate packets, worsening congestion.
- **Window Policy**: Manages the size of the sliding window at the sender side. A smaller window size limits the number of outstanding, unacknowledged packets allowed on the network at one time, naturally throttling data rates.
- **Acknowledgement Policy**: Controls when a receiver sends an ACK back to the sender. Instead of sending an ACK for every single packet, the receiver can use delayed ACKs or piggyback ACKs onto outgoing data frames, significantly reducing control traffic.
- **Discarding Policy**: Allows routers to intelligently drop packets when buffers start filling up. A smart policy drops less critical packets (like audio/video data) while preserving high-priority packets (like routing updates or control signals).
- **Admission Policy**: Used primarily in virtual-circuit networks. A router or switch will simply reject new incoming connection requests if the current traffic load threatens to degrade the quality of existing connections.

4.2.3 Approaches to Congestion Control:

Congestion control refers to the techniques and mechanisms used to manage traffic in a computer network to prevent the network from reaching a breakdown point where performance collapses.

These techniques are broadly classified into two main approaches: Open-Loop (Proactive), which focuses on preventing congestion before it happens, and Closed-Loop (Reactive), which removes congestion after it is detected.

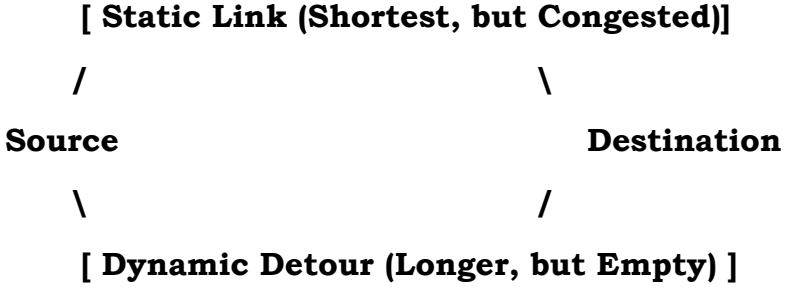
4.2.4 Traffic-Aware Routing:

Traffic-Aware Routing is a dynamic routing strategy that calculates network paths based on current traffic loads, packet delays, and queue lengths rather than static metrics like link distance or hop counts.

Unlike traditional static routing protocols (which send data along the same path regardless of how congested it is), traffic-aware routing shifts traffic away from heavily congested areas to underutilized links, optimizing overall network throughput.

Key Concepts & How It Works

Traditional routing protocols (like OSPF) often rely on static metrics, which means the shortest path is always chosen, even if that path is experiencing a severe traffic jam. Traffic-aware routing introduces dynamic metrics to solve this problem.



- Dynamic Link Weighting: The "cost" of a link is calculated dynamically using real-time factors like queuing delay, propagation delay, transmission delay, and available bandwidth.
- Early Routing Metrics: Early implementations assigned a link weight directly proportional to its queue length. However, this caused rapid oscillations, as traffic would constantly oscillate back and forth between two paths.
- Modern Metric Calculations: To prevent these oscillations, modern protocols calculate link weights based on a time-averaged smoothing function or packet delay over a fixed period rather than instantaneous queue size.

4.2.4 Admission Control:

Admission control determines whether a new traffic flow should be allowed into the network.

Example

Suppose a network is already heavily loaded.

A new video-streaming application requests network resources.

The network checks:

- Available bandwidth
- Buffer capacity
- Delay requirements
- Current traffic

If sufficient resources are unavailable:

Request is rejected or delayed.

Analogy

Think of an auditorium:

If all seats are occupied, new people are not allowed inside.

4.2.5: Traffic Throttling:

Traffic throttling means deliberately reducing the rate at which data is transmitted.

Example:

If a network normally allows:

100 Mbps

but congestion occurs, the traffic may be reduced to:

60 Mbps

Purpose

- Reduce congestion
- Prevent buffer overflow
- Improve network stability

4.2.6 Load Shedding:

Load Shedding

When congestion becomes severe, the network may deliberately discard some packets.

This is called load shedding.

Example

During extreme congestion:

Incoming packets:

P1 P2 P3 P4 P5 P6 P7 P8

↓

Router

↓

Some packets discarded

Important point

Load shedding is usually a last-resort technique.

Some applications can tolerate packet loss better than others.

For example:

- Real-time multimedia may tolerate some loss.
- Important financial transactions generally cannot.

4.3 Internet Working / Internetworking

4.3.1 What is Internetworking?

Internetworking is the process of connecting multiple heterogeneous networks so that devices on different networks can communicate.

Example:

LAN 1

|

Router

|

Internet

|

Router

|

LAN 2

The Internet itself is an example of a huge network of networks.

4.3.2 How Networks Differ

Different networks may differ in:

- Communication technology
- Addressing schemes

- Packet sizes
- Data rates
- Protocols
- Error handling
- Physical media
- Maximum transmission unit (MTU)

For example:

Ethernet Network

|

Router

|

Wireless Network

|

Router

|

Satellite Network

Internetworking protocols allow these different networks to communicate**.**

4.3.3 How Networks Can Be Connected

Networks can be connected using devices such as:

Repeater

Regenerates signals.

Bridge

Connects LAN segments.

Switch

Connects devices within a LAN.

Router

Connects different networks and selects paths.

Gateway

Connects networks using different protocols or architectures.

4.3.4 Tunnelling

Definition

Tunnelling is a technique in which one protocol packet is encapsulated inside another protocol packet to transport it across a network.

Original Packet

↓

Encapsulation

↓

+----------------------+

| Outer Header |

| Original Packet |

+----------------------+

↓

Network

↓

Decapsulation

↓

Original Packet

Example

IPv6 packets can be transported through an IPv4 network using tunneling.

IPv6 Network

|

v

IPv4 Network

|

v

IPv6 Network

4.3.4 Internetwork Routing

Internetwork routing determines how packets travel between different networks.

A router examines the destination IP address and selects the appropriate next hop.

Example:

Network A

|

Router R1

|

Router R2

|

Router R3

|

Network B

The routing table helps R1 determine where to forward the packet.

4.3.5 Fragmentation

Different networks may have different **Maximum Transmission Unit (MTU)** sizes.

Suppose a packet is:

4000 bytes

but the next network allows only:

1500 bytes

The packet may need to be divided into smaller pieces.

This process is called **fragmentation**.

Original Packet



↓

Fragmentation

↓ ↓ ↓

1500 B 1500 B 1000 B

The fragments can later be reassembled at the destination where applicable.

4.3.6 Network Layer in the Internet

The Internet Network Layer is primarily based on the **Internet Protocol (IP)**.

Its major responsibilities include:

- Logical addressing
- Routing
- Packet forwarding
- Internetworking
- Fragmentation-related functions

Two major versions are:

- **IPv4**
- **IPv6**

4.3.7 IP Protocol

IP – Internet Protocol

IP provides a mechanism for delivering packets from source to destination across interconnected networks.

Important characteristics:

- Connectionless
- Best-effort delivery
- Uses logical addresses
- Supports routing
- Does not itself guarantee reliable delivery

Important point

IP does **not** guarantee:

- Delivery
- Ordering
- Duplicate protection
- Error recovery

These functions may be handled by higher-layer protocols such as TCP.

IP Version 4 Protocol – IPv4

IPv4 is the fourth version of the Internet Protocol.

IPv4 uses a **32-bit address**.

IPv4 address

An IPv4 address consists of **32 bits**, usually represented as four decimal numbers separated by dots.

Example:

192.168.1.10

Binary representation:

11000000.10101000.00000001.00001010

Each group is called an **octet** and contains 8 bits.

Therefore:

4 octets × 8 bits = 32 bits

Number of IPv4 addresses

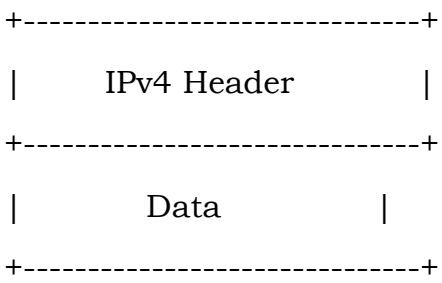
The theoretical number of addresses is:

232=4,294,967,2962^{32} = 4,294,967,296

However, not all addresses are available for assignment because some are reserved for special purposes.

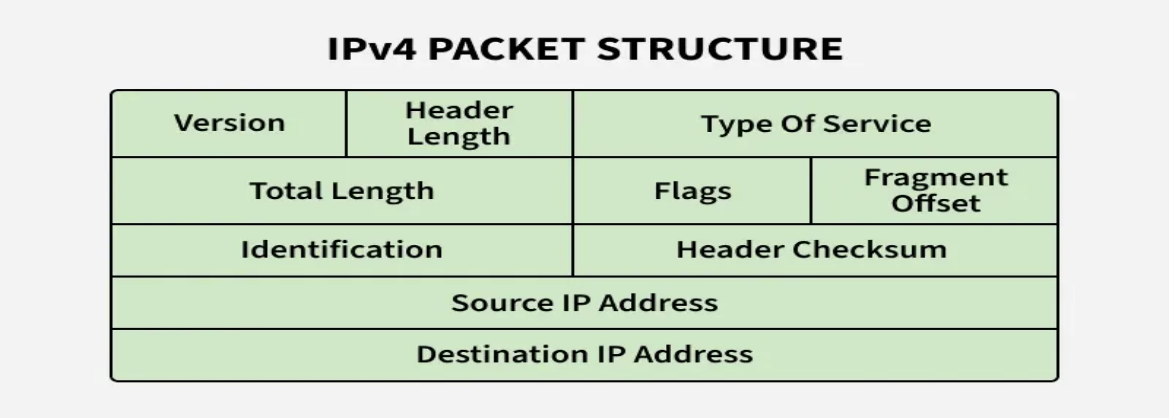
Pv4 Header Format

An IPv4 packet consists of:



The IPv4 header has a **minimum size of 20 bytes** and can be up to **60 bytes**.

IPv4 Header



Important IPv4 header fields

1. Version

Indicates the IP version.

For IPv4:

Version = 4

2. IHL

Internet Header Length specifies the length of the IPv4 header.

Minimum:

5 × 32-bit words = 20 bytes

3. Total Length

Specifies the total size of the IP packet, including header and data.

Maximum IPv4 packet size:

65,535 bytes65,535\text{ bytes}

4. Identification

Used to identify fragments belonging to the same original IP datagram.

5. Flags

Controls fragmentation.

Important flags include:

- **DF – Don't Fragment**
- **MF – More Fragments**

6. Fragment Offset

Indicates the position of a fragment within the original packet.

7. TTL

Time To Live prevents packets from circulating indefinitely.

Each router normally decreases TTL by at least one.

When TTL becomes zero, the packet is discarded.

8. Protocol

Identifies the upper-layer protocol.

Examples:

Value	Protocol
6	TCP
17	UDP
1	ICMP

9. Header Checksum

Used to detect errors in the IPv4 header.

10. Source Address

Contains the IPv4 address of the sender.

11. Destination Address

Contains the IPv4 address of the receiver.

12. Options

Optional information that can extend the header.

IP Addresses

An IP address identifies an interface on an IP network.

For IPv4, the address contains:

Network Part + Host Part

Example:

192.168.1.25

Depending on the subnet mask, part of this address identifies the network and the remaining part identifies the host.

Example

Address:

192.168.1.25

Subnet mask:

255.255.255.0

Network:

192.168.1.0

Host:

25

Classful Addressing

Originally, IPv4 addresses were divided into five classes:

Class A, B, C, D and E

Class A

Range:

1.0.0.0 – 126.255.255.255

Default mask:

255.0.0.0

Prefix:

/8

Designed for very large networks.

Class B

Range:

128.0.0.0 – 191.255.255.255

Default mask:

255.255.0.0

Prefix:

/16

Designed for medium-sized networks.

Class C

Range:

192.0.0.0 – 223.255.255.255

Default mask:

255.255.255.0

Prefix:

/24

Designed for smaller networks.

Class D

Range:

224.0.0.0 – 239.255.255.255

Used for **multicast**.

It does not have the traditional network/host division used by Classes A–C.

Class E

Range:

240.0.0.0 – 255.255.255.255

Historically reserved for experimental purposes.

Classful Addressing Summary

Class	First Octet	Default Prefix	Main Use
A	1–126	/8	Large networks
B	128–191	/16	Medium networks
C	192–223	/24	Small networks
D	224–239	—	Multicast
E	240–255	—	Experimental

Important: 0.x.x.x and 127.x.x.x have special purposes; 127.0.0.0/8 is reserved for loopback.

CIDR

CIDR = Classless Inter-Domain Routing

CIDR was introduced to overcome the limitations of classful addressing.

Instead of fixed Classes A, B and C, CIDR uses a prefix length.

Example:

192.168.10.0/24

Here:

/24 → first 24 bits are network bits

→ remaining 8 bits are host bits

Subnet mask:

255.255.255.0

CIDR example

Consider:

192.168.10.0/26

Number of host bits:

32−26=632-26=6

Number of addresses:

26=642^6=64

Usable host addresses in a traditional IPv4 subnet:

64−2=6264-2=62

The two traditionally unavailable addresses are:

- Network address
- Broadcast address

Advantages of CIDR

- Reduces IP address wastage.
- Supports flexible network sizes.
- Enables route aggregation.
- Reduces routing-table size.
- Improves IPv4 address utilization.

7. Subnets and Subnetting

Subnetting is the process of dividing one network into smaller logical networks called subnets.

Why subnetting?

Suppose an organization has:

192.168.1.0/24

It has 256 total addresses.

The organization may want separate networks for:

- Administration
- Faculty
- Students
- Laboratory

Subnetting allows the original network to be divided into smaller networks.

Example: Divide /24 into four subnets

Original network:

192.168.1.0/24

Borrow 2 host bits:

/24 → /26

Number of subnets:

22=42^2=4

Each subnet contains:

26=642^6=64

addresses.

Subnet	Network Address	Broadcast	Usable Hosts
1	192.168.1.0/26	192.168.1.63	.1 – .62
2	192.168.1.64/26	192.168.1.127	.65 – .126
3	192.168.1.128/26	192.168.1.191	.129 – .190
4	192.168.1.192/26	192.168.1.255	.193 – .254

Key formula

Number of subnets:

2n2^n

where n = number of borrowed bits.

Number of addresses per subnet:

2h2^h

where h = number of remaining host bits.

8. IP Version 6 – IPv6

IPv6 is the newer version of the Internet Protocol developed to address the limitations of IPv4, especially the shortage of IPv4 addresses.

IPv6 uses a:

128-bit address

Therefore, the theoretical address space is:

21282^{128}

which is enormously larger than IPv4's 2322^{32}.

IPv6 address example

2001:0db8:0000:0000:0000:ff00:0042:8329

IPv6 addresses are represented in hexadecimal.

They contain eight groups, with each group representing 16 bits.

IPv6 Address Shortening

Leading zeros can be removed.

Example:

2001:0db8:0000:0000:0000:ff00:0042:8329

becomes:

2001:db8:0:0:0:ff00:42:8329

Consecutive zero groups can be replaced by :: once in an address:

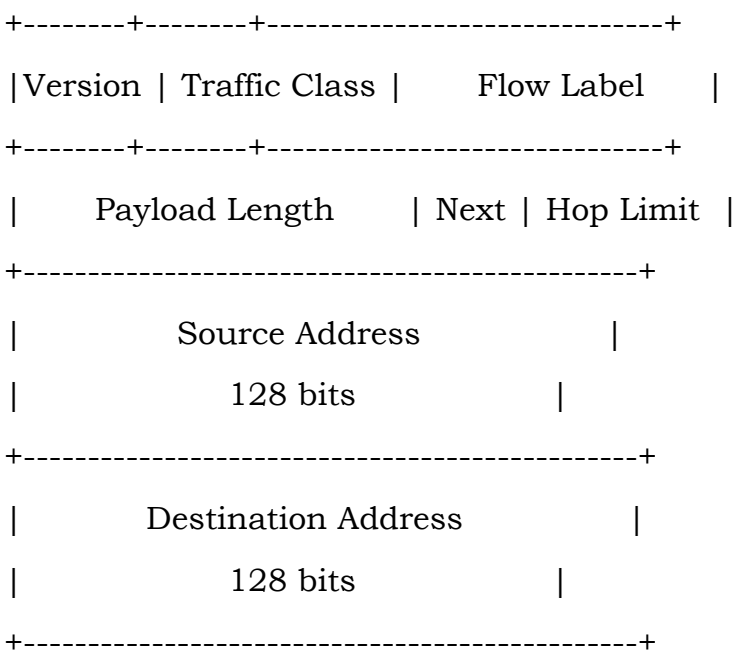
2001:db8::ff00:42:8329

---

9. Main IPv6 Header

Unlike IPv4, the IPv6 base header has a fixed size of 40 bytes.

IPv6 Header Format



Main IPv6 fields

1. Version

Specifies the IP version.

For IPv6:

Version = 6

2. Traffic Class

Used to identify traffic classes and support differentiated handling of packets.

3. Flow Label

Used to identify packets belonging to a particular flow.

4. Payload Length

Specifies the length of the payload following the IPv6 base header.

5. Next Header

Identifies the next protocol/header.

It can indicate:

- TCP
- UDP
- ICMPv6
- An IPv6 extension header

6. Hop Limit

Similar to IPv4 TTL.

Each forwarding router decreases the value.

7. Source Address

128-bit IPv6 address of the sender.

8. Destination Address

128-bit IPv6 address of the receiver.

10. IPv6 Extension Headers

IPv6 removes many optional fields from the main header and uses extension headers when additional information is required.

Examples include:

- Hop-by-Hop Options
- Routing
- Fragment
- Destination Options
- Authentication Header
- Encapsulating Security Payload

This provides a more flexible header structure.

11. Transition from IPv4 to IPv6

IPv4 and IPv6 cannot simply be switched simultaneously across the entire Internet. Therefore, several transition mechanisms are used.

Major transition approaches

1. Dual Stack

A device runs both:

IPv4 + IPv6

Example:

Computer

|

+---- IPv4

|

+---- IPv6

This allows communication with both IPv4 and IPv6 networks.

2. Tunneling

An IPv6 packet is carried through an IPv4 network.

IPv6 Network

|

| IPv6 packet

↓

IPv4 Network

|

| Encapsulated IPv6 packet

↓

IPv6 Network

The IPv6 packet is encapsulated inside an IPv4 packet while crossing the IPv4 network.

3. Translation

A translation mechanism allows communication between IPv6-only and IPv4-only systems by translating between the protocols.

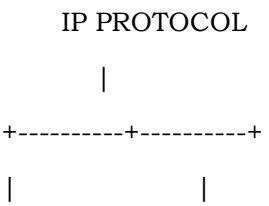
A commonly used approach is NAT64, often used together with DNS64.

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12. Comparison of IPv4 and IPv6

Feature	IPv4	IPv6
Address size	32 bits	128 bits
Address notation	Decimal	Hexadecimal
Example	192.168.1.10	2001:db8::1
Header size	20–60 bytes	40 bytes base header
Header checksum	Present	Removed from base header
Broadcast	Supported	No broadcast
Multicast	Supported	Supported
Fragmentation	Hosts and routers can fragment	Fragmentation is performed by source using Fragment extension header
Address configuration	Manual/DHCP	SLAAC, DHCPv6, manual
TTL equivalent	TTL	Hop Limit
IPsec	Defined for IPv4 but historically optional	Support standardized in IPv6 architecture
Address exhaustion	Major limitation	Vast address space
Header options	IPv4 options	Extension headers
ARP	Used	Neighbor Discovery (NDP)
QoS-related field	ToS/DS field	Traffic Class + Flow Label

13. Easy Conceptual Flow :



IPv4 IPv6

| |

32-bit address 128-bit address

| |

IPv4 Header IPv6 Header

| |

Classful Addressing Extension Headers

|

CIDR

|

Subnetting

|

IPv4 → IPv6 Transition

|

Dual Stack

Tunneling

Translation

****

Important points for examination

2-mark questions

1. Define IP protocol.
2. What is IPv4?
3. What is the size of an IPv4 address?
4. What is the purpose of TTL?
5. What is CIDR?
6. Define subnetting.
7. What is an IPv6 address?
8. What is the size of an IPv6 address?
9. What is the purpose of the IPv6 Flow Label?
10. What is Dual Stack?

11. What is IP tunneling?
12. State two differences between IPv4 and IPv6.

5/10-mark questions

1. Explain the IPv4 header format with a neat diagram.
2. Explain IPv4 classful addressing with Classes A, B, C, D and E.
3. Explain CIDR with a suitable example.
4. Explain subnetting and solve a subnetting example.
5. Explain the main IPv6 header with a neat diagram.
6. Explain IPv4-to-IPv6 transition mechanisms.
7. Compare IPv4 and IPv6 in detail.
8. Explain IPv6 extension headers and their purpose.

****